

- 6 Figure 2.34 shows an arrangement used to measure the acceleration of a metal plate as it falls vertically. The metal plate is released from rest and falls a distance of 0.200 m before breaking light beam 1. It then falls a further 0.250 m before breaking light beam 2.

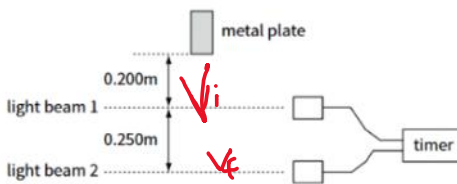


Figure 2.34 For End-of-chapter Question 6.

- a Calculate the time taken for the plate to fall 0.200 m from rest. (You may assume that the metal plate falls with an acceleration equal to the acceleration of free fall.)
- b The timer measures the speed of the metal plate as it falls through each light beam. The speed as it falls through light beam 1 is 2.2 m s^{-1} and the speed as it falls through light beam 2 is 3.2 m s^{-1} .
- i Calculate the acceleration of the plate between the two light beams.
- ii State and explain one reason why the acceleration of the plate is not equal to the acceleration of free fall.

a)

$$s = \frac{gt^2}{2} \Rightarrow 0.4 = 10t^2$$

$$v^2 = v_i^2 + 2as$$

v_i, v_f

time to reach to light beam 1: $t = 0.2 \text{ s}$

$v_0 = 0$ (free fall)

$$v_i^2 = (eq. 4) \quad v_0^2 + 2gS_0$$

$$v_i = gt$$

equation 1:	$v = u + at$
equation 2:	$s = \frac{(u+v)}{2} \times t$
equation 3:	$s = ut + \frac{1}{2}at^2$
equation 4:	$v^2 = u^2 + 2as$

- 10 A hot-air balloon rises vertically. At time $t = 0$, a ball is released from the balloon. Figure 2.38 shows the variation of the ball's velocity v with t . The ball hits the ground at $t = 4.1 \text{ s}$.

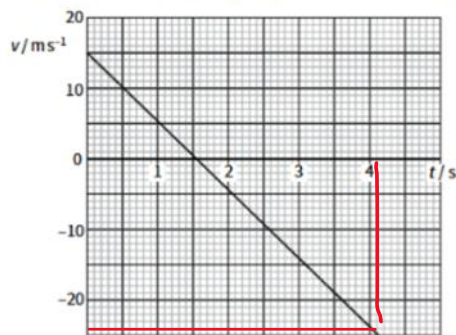
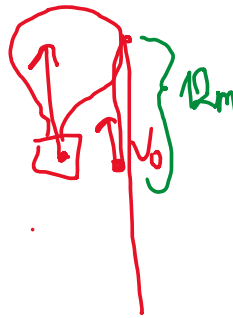


Figure 2.38 For End-of-chapter Question 10.

- a Explain how the graph shows that the acceleration of the ball is constant.
- b Use the graph to:
- determine the time at which the ball reaches its highest point
 - show that the ball rises for a further 12 m between release and its highest point
 - determine the distance between the highest point reached by the ball and the ground.
- c The equation relating v and t is $v = 15 - 9.81t$. Explain the significance in the equation of:
- the number 15
 - the negative sign.

c)



Initial velocity at $t=0$
 $V_0 = 15 \text{ m/s}$

i) Velocity at the highest point: $V_{\text{highest}} = 0$

$\Rightarrow t = 1.5 \text{ s}$

ii)

$$S = v_0 t - \frac{1}{2}gt^2$$

$$S = 15 \cdot 1.5 - 5 \cdot 2.25 =$$

iii)

$$t = 4.1$$

$v =$

Time needed to fly from the highest point to the ground:

$$t = 4.1 - 1.5 = 2.6$$

$$V_{\text{highest}} = 0$$

$$S = V_{\text{highest}} t - \frac{1}{2}gt^2 = \frac{9.8 \cdot 2.6^2}{2} = 33.1$$

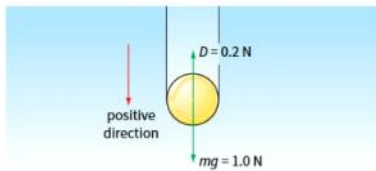
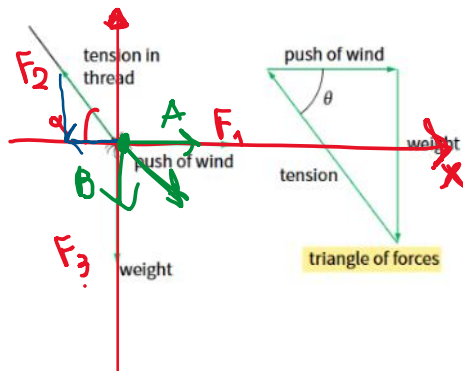
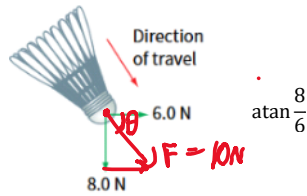
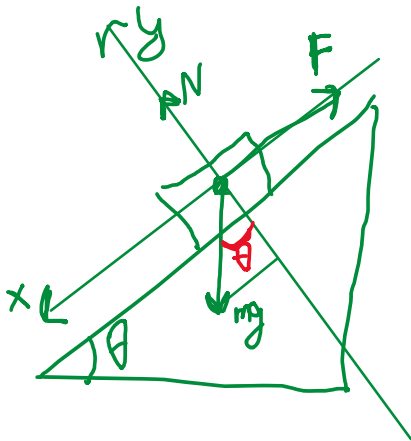


Figure 4.3 Two forces on a falling tennis ball.



$$\begin{aligned} X: F_{1x} &= F_1 \\ F_{3x} &= 0 \\ F_{2x} &= F_2 \cos \alpha \\ F_{netx} &= F_1 - F_2 \cos \alpha = A \\ F_{nety} &= B \end{aligned}$$



- 1 A boy of mass 40 kg is on a waterslide which slopes at 30° to the horizontal. The frictional force up the slope is 120 N. Calculate the boy's acceleration down the slope. Take the acceleration of free fall g to be 9.81 ms^{-2} .

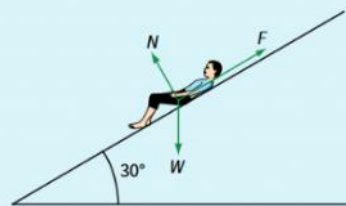


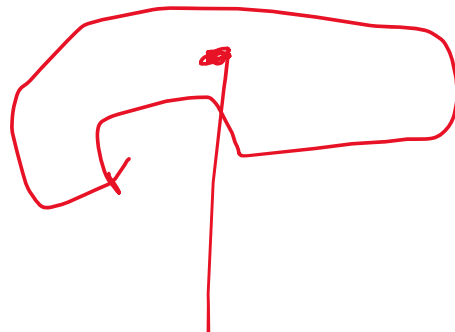
Figure 4.13 For Worked example 1.

$$\sum F_{ix} = ma_x$$

$$\sum F_{iy} = ma_y = 0$$

$$\begin{aligned} X: \\ mg_x &= mg \sin \theta \\ N_x &= 0 \\ F_x &= -F \\ mg \sin \theta - F &= ma_x \end{aligned}$$

$$\begin{aligned} Y: \\ mg_y &= -mg \cos \theta \\ N_y &= N \\ F_y &= 0 \\ N - mg \cos \theta &= 0 \\ N &= mg \cos \theta \end{aligned}$$



The centre of gravity of an object is defined as the point where all the weight of the object may be considered to act.

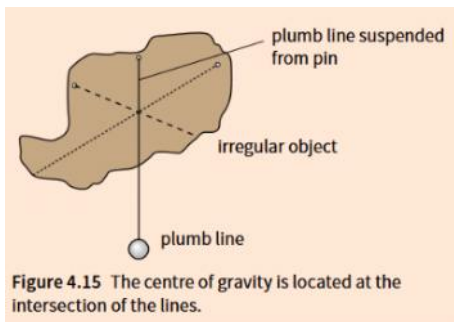


Figure 4.15 The centre of gravity is located at the intersection of the lines.

Homework:

- 5 A crate is sliding down a slope. The weight of the crate is 500 N. The slope makes an angle of 30° with the horizontal.
 - a Draw a diagram to show the situation. Include arrows to represent the weight of the crate and the contact force of the slope acting on the crate.
 - b Calculate the component of the weight down the slope.
 - c Explain why the contact force of the slope has no component down the slope.
 - d What third force might act to oppose the motion? In which direction would it act?

- 1 A ship is pulled at a constant speed by two small boats, A and B, as shown in Figure 4.27. The engine of the ship does not produce any force.

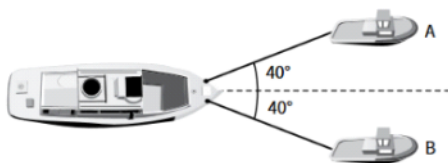


Figure 4.27 For End-of-chapter Question 1.

The tension in each cable between A and B and the ship is 4000 N.

- a Draw a free-body diagram showing the three horizontal forces acting on the ship.
 - b Draw a vector diagram to scale showing these three forces and use your diagram to find the value of the drag force on the ship.
- 2 A block of mass 1.5 kg is at rest on a rough surface which is inclined at 20° to the horizontal as shown in Figure 4.28.

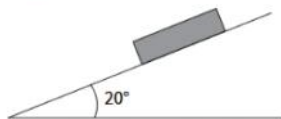


Figure 4.28 For End-of-chapter Question 2.

- a Draw a free-body diagram showing the three forces acting on the block.
- b Calculate the component of the weight that acts down the slope.
- c Use your answer to b to determine the force of friction that acts on the block.
- d Determine the normal contact force between the block and the surface.

- 3 The free-body diagram (Figure 4.29) shows three forces that act on a stone hanging at rest from two strings.

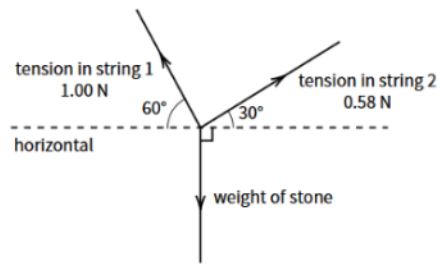


Figure 4.29 For End-of-chapter Question 3.

- Calculate the horizontal component of the tension in each string. Why should these two components be equal in magnitude?
- Calculate the vertical component of the tension in each string.
- Use your answer to b to calculate the weight of the stone.
- Draw a vector diagram of the forces on the stone. This should be a triangle of forces.
- Use your diagram in d to calculate the weight of the stone.